

CONTACT INFORMATION	linkedin.com/in/aarthi-venkat/ aarthivenkat.github.io	(408) 799-9189 avenkat@broadinstitute.org
EDUCATION	Yale University Ph.D. in Computational Biology & Bioinformatics M.S. in Computational Biology & Bioinformatics The University of California, San Diego B.S. in Bioengineering: Bioinformatics	May 2024 Dec 2021 Jun 2019
RESEARCH EXPERIENCE	Postdoctoral Associate Eric and Wendy Schmidt Center, Broad Institute of MIT and Harvard & Harvard Medical School, Drs. Nir Hacohen & Marinka Zitnik - Build foundation and graph-based AI models for functional genomics in systems immunology, integrating diverse datasets to generate actionable and testable insights Computational Biology & Bioinformatics Ph.D. Yale University, Dr. Smita Krishnaswamy - Developed 7+ geometric representation learning methods for single-cell and spatial sequencing data to reveal underlying cellular and molecular behavior - Co-led 8+ collaborations across immunology, cancer, and genetics for characterization and target/biomarker discovery across multiple biological scales and modalities Applied Science Research Intern Google Brain, Drs. Lucy Colwell & Farhad Hormozdiari - Performed ML-guided biological sequence design with Google Genomics, UCSF - Achieved top performance for CRISPR RNA guide efficacy and expression prediction Bioinformatics Research Assistant La Jolla Institute for Immunology, Dr. Ferhat Ay - Characterized 3D structure of malaria-related parasite genomes from Hi-C sequencing - Corrected <i>Toxoplasma gondii</i> misassembly with Hi-C and long-read sequencing Computational Biology Research Assistant Institute for Genomic Medicine, Dr. Theresa Gaasterland - Performed bioinformatic analysis of primary congenital glaucoma exomes Genome Informatics Intern Regeneron Genetics Center, Regeneron Pharmaceuticals - Integrated loss-of-function variant and target annotation for over 500,000 exomes - One of 7 selected out of 250+ interns to present at company-wide annual event Data Analytics Intern Research & Development, Auris Health - Built cloud-based pipeline to facilitate high-performance analysis of endoscopic robot	Sept 2024-Present Aug 2019-Jul 2024 Sept 2021-Dec 2021 Oct 2016-Sept 2019 Sept 2018-Aug 2019 Jun 2018-Sept 2018 Jun 2017-Sept 2017
SELECTED PUBLICATIONS	*§ Equal contribution. Links to full articles at: https://aarthivenkat.github.io	

[1] **A Venkat**, Z Jiang, D Marbach, N Hacohen[§], M Zitnik[§]. *Cross-modal transfer learning for mapping bulk transcriptomes at cellular level* (2026).
– Presented at ICML 2026 3rd Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences, Eric and Wendy Schmidt Symposium 2026: Biomedical Science and AI

- [2] C Garcia*, **A Venkat***, DC McQuaid* ... S Krishnaswamy[§], MD Muzumdar[§]. *Beta cell-derived cholecystokinin drives obesity-associated pancreatic adenocarcinoma development*. Nature Communications (2026).
 – Presented at WNAR/IMS 2026, ISMB MLCSB 2025, AACR 2022/2024/2025, Yale Single Cell Symposium 2022, Broad Institute Symposium 2024, Brown University Nabrit Conference 2025
 – Followed up by McQuaid et al. (AACR 2026, effect of treatment with GLP1-RAs)
- [3] **A Venkat***, S Youlten*, BP San Juan* ... S Krishnaswamy[§], CL Chaffer[§]. *AAnet resolves a continuum of spatially-localized cell states to unveil tumor complexity*. Cancer Discovery (2025).
 – Featured selection by AACR Publications at 2026 AACR Meeting
 – Presented at WNAR/IMS 2026, UoI Math & Statistics Colloquium 2026, CSHL Systems Immunology 2021
 – Followed up by Garcia, Venkat, McQuaid et al. (Nature Communications, beta cell archetypes), Torices et al. (In progress, atlas-level archetypal analysis)
- [4] **A Venkat**, S Leone, S Youlten, E Fagerberg, J Attanasio, NS Joshi, S Krishnaswamy. *Mapping the gene space at single-cell resolution with gene signal pattern analysis*. Nature Computational Science (2024).
 – Presented at UoI Math & Statistics Colloquium 2026, Learning on Graphs 2025, Yale Genetics Symposium 2023, Gruber Science Symposium 2023, AAI 2023, ISMB MLCSB 2021
 – Followed up by Thomas et al. in trainee-led work (ASCB 2025, patient-specific gene-gene networks)

JOURNAL ARTICLES

- [5] **A Venkat***, M Carlino*, B Lawton* ... S Krishnaswamy[§], D Krause[§]. *Single-cell analysis reveals transcriptional dynamics in primary parathyroid tissue*. Genome Research (2024).
- [6] **A Venkat**, D Bhaskar, S Krishnaswamy. *Multiscale geometric and topological analyses for characterizing and predicting immune responses from single-cell data*. Cell Trends in Immunology (2023).
- [7] M Damo, N Hornick, **A Venkat** ... NS Joshi. *PD-1 prevents pathogenicity of effector CD8 T cells that infiltrate skin under homeostatic conditions*. Nature (2023).
- [8] M Amodio, SE Youlten, **A Venkat**, BP San Juan, CL Chaffer, S Krishnaswamy. *Single-cell multi-modal GAN reveals spatial patterns in single-cell data from triple-negative breast cancer*. Cell Patterns (2022).
- [9] KA Connolly, M Kuchroo, **A Venkat** ... NS Joshi. *A reservoir of stem-like CD8+ T cells in the tumor draining lymph node preserves the ongoing antitumor immune response*. Science Immunology (2021).
- [10] Y Su*, **A Venkat***, Y Yadav, L Puglisi, S Fodeh. *Twitter-based analysis reveals differential COVID-19 concerns across areas with socioeconomic disparities*. CBM (2021).
- [11] J Xia, **A Venkat**, ML Reese, KG Le Roch, F Ay, JP Boyle. *Third generation sequencing revises the molecular karyotype for Toxoplasma gondii and identifies emerging copy number variants in sexual recombinants*. Genome Research (2021).
- [12] EM Bunnik, **A Venkat***, J Shao*, KE McGovern ... F Ay[§], KG Le Roch[§]. *Comparative 3D Organization in Apicomplexan Parasites*. PNAS (2019).

CONFERENCE PROCEEDINGS & WORKSHOP PAPERS

- [13] **A Venkat***, J Chew*, F Cardoso Rodriguez, CJ Tape, M Perlmutter[§], S Krishnaswamy[§]. *Directed scattering for knowledge graph-based cellular signaling analysis*. ICASSP (2024).
- [14] D Bhaskar*, DS Magruder*, M Morales, E De Brouwer, **A Venkat**, F Wenkel, J Noonan, G Wolf, N Ivanova, S Krishnaswamy. *Inferring dynamic regulatory interaction graphs from time series data with perturbations*. LoG Conference (2023).

- [15] S Leone, A Tong, G Huguet, **A Venkat**, G Wolf, S Krishnaswamy. *Graph Fourier MMD for Signals on Graphs*. SampTA (2023).

PREPRINTS &
IN REVIEW

- [16] X Sun*, S Gupta*, A Tong*, M Kuchroo*, D Bhaskar*, C Liu, **A Venkat** ... CL Chaffer[§], S Krishnaswamy[§]. *Revealing dynamic temporal trajectories and underlying regulatory networks with Cflows*. In Review.

PRESENTATIONS

- *Cross-modal transfer learning for mapping bulk transcriptomes at cellular level*, ICML 3rd Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences (2026)
- *Uncovering cellular mechanisms of tumorigenesis with archetypal analysis*
Invited talk, WNAR/IMS Annual Meeting (2026)
- *Cross-modal transfer learning for mapping bulk cancer phenotypes at cellular resolution*, EWSC Symposium: Biomedical Science and AI Poster Presentation (2026)
- *Semantic Analysis Reveals Increasing Attribution of Responsibility to AI in Scientific Literature*, WIBI Oral & Poster Presentation (2026)
- *Relational ethics for scientific knowledge production and the limits of LLMs*
Machine Logos: Persons, Language, and AI, Oral Presentation (2026)
- *Decoding cellular biology with geometric machine learning*
Invited talk, University of Idaho Math & Statistics Colloquium (2026)
- *Analysis of Therapeutic Response in Hepatocellular Carcinoma with Graph-based Machine Learning Methods on Spatial Transcriptomics Data*
ASCB Poster Presentation (2025)
- *Mapping the gene space at single-cell resolution with gene signal pattern analysis*
LOG Poster Presentation (2025)
- *Epistemic Responsibility and Interdisciplinarity in the Age of AI for Science*
LLMs and Digital Autonomy: From Misinformation to Context Collapse Oral Presentation (2025)
- *Dissecting cellular and molecular mechanisms of pancreatic cancer with deep learning*
ISMB MLCSB Conference Oral Presentation (2025)
- *Dissecting cellular and molecular mechanisms of pancreatic cancer*
Dr. Samuel M. Nabrit Conference, Brown University, Oral Presentation (2025)
- *Geometric representation learning for single-cell biology across contexts*
Invited talk, Brown University (2025)
- *Learning cellular and molecular mechanisms of pancreatic cancer*
Connecting the Dots Broad Institute Symposium Poster Presentation (2024)
- *Mapping the gene space at single-cell resolution with gene signal pattern analysis*
Yale Department of Genetics Symposium Poster Presentation (2023)
- *Mapping the gene space at single-cell resolution with gene signal pattern analysis*
Gruber Science Fellowship Symposium Poster Presentation (2023)
- *Learning directed and hyperbolic embeddings*
Graph Signal Processing Workshop Oral Presentation (2023)
- *PHATE reveals cell state transformation in Tercen biomedical data analysis platform*
CYTO Oral Presentation (2023)
- *Mapping the gene space at single-cell resolution with gene signal pattern analysis*
AAI Immunology Poster Presentation (2023)
- *Elucidating mechanisms of endocrine-exocrine signaling in pancreatic cancer*
Yale Single Cell Symposium Oral Presentation (2022)
- *Manifold-based gene density estimates reveal immune signaling in meningioma*
ISMB MLCSB Conference Poster Presentation (2021)
- *Archetypal analysis of antigen-specific T cell responses across conditions*
CSHL Systems Immunology Conference Poster Presentation (2021)

- *Leveraging the Power of Human Genetics through Knockout Discovery*
Regeneron Oral Presentation & Poster (2018)

TEACHING EXPERIENCE	Teaching Assistant, Computational Genomics	Nov 2022, Dec 2023
	Cold Springs Harbor Laboratory Workshop	
	• Designed and presented single-cell workshops for 20-40 PhD-level researchers	
	Teaching Fellow, Deep Learning Theory and Applications	S 2021, S 2024
	Yale University, Computer Science	
	• Held recitations, designed and graded homework, exams, and projects for undergraduate and graduate students	
	Teaching Assistant, Machine Learning for Single-cell Analysis	May 2020, Jan 2021
	Yale University, Department of Genetics & Yale SEAS	
	• Co-taught 100+ researchers across all levels in tools for single-cell analysis	
	Teaching Assistant, Introduction to Biomedical Data Science and Health Informatics	Jun 2020
	Yale Center for Medical Informatics	
	• Assisted in Python for biomedical data analysis for researchers across all levels	
	Genetics Undergraduate Tutor / Instructional Assistant	F 2017, F 2018, S 2019
	UC San Diego Biological Sciences	
	• Developed material for weekly recitation sessions, office hours, and exam preparation • Received Excellence in Teaching Award for top performance (100% positive reviews)	
FELLOWSHIPS & GRANTS	GSA Conference Travel Fellowship	Jun 2023, Mar 2024
	Yale University	
	Yale Gruber Science Fellowship	Aug 2019
	Yale University	
	• Most prestigious award offered by Graduate School of Arts and Sciences to incoming science PhDs in recognition of outstanding accomplishments and promise	
HONORS & AWARDS	Public Communication Certificate	2023
	Poorvu Center for Teaching & Learning, Yale University	
	• Certificate for skills developed in oral and written communication	
	OHHER Award Finalist for Yale Research Excellence	2022
	Yale School of Medicine, Office of Health Equity Research	
	• Received for “Twitter-based analysis reveals differential COVID-19 concerns across areas with socioeconomic disparities”	
	Outstanding Academic Achievement in Bioengineering	2019
	The University of California, San Diego	
	• Highest performance in graduating class in Bioengineering: Bioinformatics	
	Excellence in Teaching Award	2019
	The University of California, San Diego	
	• Highest performance evaluation for teaching assistance in Genetics	
	Tau Beta Pi Engineering Honors	2018, 2019
	The University of California, San Diego	
	• Awarded to engineering students displaying high academic achievement and personal, professional integrity	
	Muir College Caledonian Honors	2018, 2019
	The University of California, San Diego	
	• Awarded to engineering students displaying high academic achievement and personal, professional integrity	

ACADEMIC SERVICE	Provost Honors	2015-2019
	The University of California, San Diego	
	Received 12 times for high academic achievement	
	Invited Reviewer and PC Member	2023-2026
	- Reviewed and co-reviewed papers for Nature, BMC Bioinformatics, Computational & Structural Biotechnology Journal, Yale Journal of Biology & Medicine - PC (ICML GFM 2026, ICML GenBio 2026, ICLR Gen² 2026, ISMB 2026, ISMB 2025), Subreviewer (RECOMB 2024)	
	Research Mentor	2023-2026
	- Hannah Thomas, Nashua High School South → Research Science Institute @ MIT - Sofia Lara, MIT Biological Engineering → Rhodes Scholar PhD @ Oxford University - Sam Leone, Yale Applied Mathematics → MIT Lincoln Laboratory	
	KeyPals Volunteer	2026
	Provide mentorship and writing/computer skills for local Cambridge fifth graders	
	Broad Summer Research Program Selection Committee	2025
	Evaluate applicants for BSRP program for underrepresented students with a strong interest in genomics research	
	Foundation Models in Genomics Panel Moderator @ Broad	2024
	Led discussion on opportunities, successes, and pitfalls of recent genomic and transcriptomic foundation models	
	Student Advisory Board, Poorvu Center for Teaching & Learning	2023
	Developed curriculum and policy incorporating AI literacy and DEI principles	
	Networking Chair, Yale Gruber Science Fellowship	2022, 2023
	Hosted networking talks, panels, and discussion to foster Gruber scientific community	
	Student Representative, Graduate Student Assembly	2023
	Advocated for CB&B graduate students to improve Yale healthcare literacy and policy	
	Reviewing Editor, Yale Journal of Biology & Medicine	2023
	Managed <i>Big Data</i> issue manuscripts, inviting reviewers and making editorial decisions	
	Social Services & Insurance Counseling, HAVEN Free Clinic	2022, 2023
	Provided healthcare guidance and resources to uninsured New Haven residents	
	Cancer Biology Training Program, Yale School of Medicine	2021-2023
	- Completed certificate in cancer biology through additional translational coursework - Shadowed Dr. Pamela Kunz and discussed clinical relevance of <i>in silico</i> cancer research	